

Shape-Sensing Robotic-Assisted Bronchoscopic Cryoablation for Primary and Metastatic Pulmonary Nodules: Feasibility, Safety, and Early Outcomes

Liqin Xu^{a,b} Wenduo Gu^a Mitchell Zhao^c Grace Lin^c
Kerian Van Nostrand^a Niraj M. Patel^a Bryce Duchman^a
Brian Daniel Tran^a Jorge A. Munoz Pineda^a Russell Jason Miller^{a,d}
Felix J.F. Herth^{e,f} George Cheng^a

^aDepartment of Interventional Pulmonology, Bronchoscopy, and Pleural Diseases, Division of Pulmonary, Critical Care and Sleep Medicine, UC San Diego Health, La Jolla, CA, USA; ^bDepartment of Pulmonary and Critical Care Medicine, Affiliated Hospital of Nantong University, Nantong, China; ^cDepartment of Pathology, University of California, San Diego, La Jolla, CA, USA; ^dDepartment of Pulmonary Medicine, Naval Medical Center San Diego, San Diego, CA, USA; ^eDepartment of Pneumology and Critical Care Medicine, Thoraxklinik University of Heidelberg, Heidelberg, Germany; ^fTranslational Lung Research Center Heidelberg, Germany Lung Research Foundation (DZL), Heidelberg, Germany

Keywords

Shape-sensing robotic-assisted bronchoscopy ·
Transbronchial cryoablation · Pulmonary nodules ·
Feasibility · Safety · Early outcomes

Abstract

Introduction: Minimally invasive bronchoscopic interventions are increasingly explored as alternatives to surgery for pulmonary nodules. Clinical evidence on transbronchial cryoablation guided by robotic-assisted bronchoscopy (RAB) and cone-beam computed tomography (CBCT) remains limited. This study evaluated the feasibility, safety, and long-term effectiveness of shape-sensing RAB (ssRAB)-guided transbronchial cryoablation for lung nodules, including metastatic lesions. **Methods:** We performed a retrospective review of patients who underwent ssRAB-guided transbronchial cryoablation using ERBECRYO-

2 cryotherapy system between July 2023 and May 2025 at the University of California, San Diego. Feasibility was defined as successful probe positioning and completion of cryoablation; safety was assessed by procedure-related adverse events within 14 days, and efficacy was defined as radiographic local control during serial follow-up. Histopathologic findings from nodules subsequently resected were recorded. **Results:** Fourteen patients with 15 nodules (mean age, 70.3 ± 8.6 years; mean nodule size, 11.1 ± 4.9 mm) were included. ssRAB-guided cryoablation was successfully performed in all cases (feasibility 100%). No procedure-related serious adverse events occurred; minor adverse events included throat irritation (21.4%), dyspnea (14.3%), and isolated hemoptysis, pleuritic pain, or pulmonary infection (7.1% each). The median follow-up time after the transbronchial cryoablation is 110 days, with a range, 13–487 days. Local control was achieved in 86.7% of nodules. Eight nodules were resected after a

median of 79 days, showing fibrosis and inflammation on pathology. Seven nodules were followed radiographically for a median of 358 days, with 85.7% (6/7) remaining stable. Patients receiving immunotherapy demonstrated a trend toward longer local progression-free survival. **Conclusion:** ssRAB transbronchial cryoablation using a CO₂-based Erbe system is feasible and demonstrates a favorable safety profile with encouraging early local control across primary and metastatic nodules. The approach enables precise, intraprocedural confirmation of probe placement and may serve as an adjunct to systemic therapies. Prospective multicenter studies are needed to validate these findings, refine cryoablation protocols, establish radiographic and pathologic markers of complete ablation, and clarify the role of transbronchial cryoablation within multidisciplinary lung cancer care.

© 2026 The Author(s).
Published by S. Karger AG, Basel

Introduction

Pulmonary nodule management encompasses spectrum of strategies from active surveillance to surgical resection and minimally invasive local therapies. Surgical resection remains the recommended treatment for localized lung tumors, oligometastatic, and solitary lung metastasis; however, a substantial portion of patients are medically inoperable or at high surgical risk due to comorbidities [1, 2]. For these patients – and for those seeking lung-preserving approaches – minimally invasive local therapies such as stereotactic body radiation therapy (SBRT) and percutaneous ablation have emerged as effective alternatives [3, 4]. Nonetheless, these modalities are not without risks, with potential complications including radiation pneumonitis, pneumonia, pneumothorax, pleural effusion, and hemothorax [5].

Transbronchial ablation modalities have emerged as a promising extension of minimally invasive lung nodule management, with early reports suggesting a favorable safety profile and the potential to achieve local nodule control while integrating diagnosis, staging, and therapy within a single bronchoscopic procedure [6–9]. Techniques such as radiofrequency [6], microwave [8], or cryotherapy [10–13] demonstrated feasibility and encouraging short-term outcomes in selected patient cohorts. Among these, transbronchial cryoablation has garnered particular interest for its potential immunomodulatory effects [14], supported by preclinical swine models [11, 12], isolated case reports [15], and early human feasibility studies [10, 13]. A recent first-in-human phase I trial evaluated the safety and feasibility

of bronchoscopic cryoimmunotherapy delivered during standard-of-care bronchoscopy in patients with advanced, inoperable NSCLC, and reported exploratory evidence of systemic immune modulation based on peripheral blood immune and cytokine analyses [16]. Historically, however, bronchoscopic cryotherapy has been primarily applied in the central airway domain (e.g., tumor ablation/debulking, airway palliation, cryobiopsy), predominantly using CO₂-based flexible cryoprobes such as the Erbe platform [17]. The extension of this well-established bronchoscopic cryotherapy system into parenchymal targets such as peripheral pulmonary nodules remains poorly defined in the literature. Therefore, further investigation is needed to clarify its oncologic efficacy, optimal technique, and long-term clinical impact.

Robotic-assisted bronchoscopy (RAB), combined with CBCT, offers precise navigation to nodules, real-time imaging, stable robotic control and accurate access to peripheral pulmonary nodules [18]. Early studies have shown high localization success and low complication rates with robotic bronchoscopy-guided interventions [19]. Integrating RAB with transbronchial cryoablation offers new opportunities for precise and minimally invasive nodule ablation [6] and may further improve procedural outcomes and patient safety.

To our knowledge, no published human series have specifically evaluated transbronchial cryoablation of peripheral pulmonary nodules using the CO₂-based Erbe system, despite its widespread use for central airway cryotherapy and cryobiopsy. Accordingly, extending this familiar bronchoscopic cryotherapy platform to parenchymal targets has emerged as an area of growing interest.

In this study, we retrospectively review our institutional using shape-sensing RAB with cone-beam CT (CBCT) confirmation to guide transbronchial cryoablation of primary and metastatic pulmonary nodules with the ERBECRYO-2 system. Our objective was to evaluate the feasibility, safety, and early oncologic performance of this technique and to provide initial clinical insights into its potential role within multimodality lung cancer management.

Methods

Study Design

We conducted a retrospective, single-center study evaluating the feasibility, safety, and early effectiveness of transbronchial cryoablation of lung nodules performed

at the University of California San Diego (UC San Diego) Interventional Pulmonology Program. All procedures occurred between July 2023 and May 2025. The study was approved by the UC San Diego Institutional Review Board (Approval number: #812977) and conducted in accordance to the Declaration of Helsinki (revised in 2013). Written informed consent was obtained from all patients.

The primary objective was to assess technical feasibility and procedural safety. The secondary objective was to evaluate local tumor control during follow-up.

Patients

This study included all consecutive patients who underwent shape-sensing RAB (ssRAB)-guided transbronchial cryoablation using the ERBECRYO-2 CO₂-based cryotherapy system at our institution during the study period. The decision to proceed with cryoablation was made through a multidisciplinary discussion involving the primary oncology team and the patient. All patients who underwent bronchoscopic cryoablation were included in the analysis. Cases were identified retrospectively from procedural logs and electronic medical records. Baseline data collected included demographics (age, sex), smoking history, comorbidities, concurrent or prior immunotherapy, and detailed nodule characteristics (location, size, volume, density, and histology). Nodules were categorized as primary lung cancer, metastatic lesions, or benign/inflammatory based on cytopathologic confirmation obtained by rapid on-site evaluation.

Cryoablation Technique

Pre-procedural airway planning was performed using the Ion™ robotic bronchoscopy system (Intuitive Surgical, Sunnyvale, CA, USA) with integration of a mobile CBCT platform (Siemens, Malvern, PA, USA). During the procedure, target lesions were navigated to using image-guided robotic bronchoscopy, with intralesional confirmation achieved by radial endobronchial ultrasound (EBUS) and/or CBCT imaging. Probe position was adjusted in real time as needed based on imaging feedback to ensure accurate lesion targeting. In one case, the pulmonary nodule was initially localized and diagnosed using ssRAB. Due to its proximity to the central airway, cryoablation was subsequently performed under linear EBUS guidance to allow safer probe positioning.

In all cases, diagnostic biopsy and transbronchial cryoablation were performed during the same bronchoscopic session. Cytopathologic confirmation was obtained by rapid on-site evaluation, and cryoablation

was undertaken only after on-site pathologic confirmation of diagnosis.

Cryoablation was performed using the flexible cryoprobe (diameter 1.7 mm, ERBE, Tübingen, Germany) with carbon dioxide meeting United States Pharmacopeia (USP) standards as the cryogen. Freeze-thaw cycles were individualized based on lesion size, anatomic considerations, and probe stability, and were determined in real time by a team of experienced interventional pulmonologists ($n = 4$). Total cryoablation time and mode were recorded.

Hemostasis during the procedure was maintained by applying iced saline lavage and/or topical epinephrine (1:10,000), as clinically indicated. Patients were monitored post-procedure for 24 h in the hospital.

Feasibility and Safety of Transbronchial Cryoablation

Feasibility was defined as accurate ssRAB-guided probe placement confirmed by CBCT and successful completion of the planned cryoablation. The technical safety was evaluated with the incidence of procedure-related adverse events after (within 14 days) the transbronchial cryoablation. Adverse events were categorized using the Common Terminology Criteria for Adverse Events (CTCAE), version 6. All complications were reviewed by an interventional pulmonologist independent of the procedural team to determine relevance to the procedure. Specific adverse events included peri-procedural pneumothorax, pulmonary infection, bloody sputum/hemoptysis, pleural effusion, shortness of breath, chest pain/pleural pain.

Effectiveness and Follow-Up

All patients had chest CT and/or PET/CT prior to ablation. For follow-up, patients were evaluated with CT post-procedure after 1 day to 487 days (Table 1). Local tumor control was assessed at the lesion level on follow-up CT or PET/CT. Local progression was defined as new or enlarging nodular enhancement (diameter increase by 20%) within or adjacent to the ablation zone compared to the first post-ablation CT, or an increase in metabolic activity on PET not attributable to post-ablation inflammation. For nodules with serial imaging, the maximal long-axis (DL) and perpendicular maximum short-axis diameter (DS) of the ablation zone were measured, and calculate the ablation volume ($\pi/6 \times DL \times DS^2$) was recorded.

Surgical Cohort

Decisions regarding subsequent surgical resection versus radiologic surveillance were made based on multidisciplinary clinical assessment, considering tumor

Table 1. Patient demographics and nodule characteristics

Characteristics	Value
Patients, <i>n</i>	14
No. of nodules	15
Age, years	70.3±8.6 [51.0–81.0]
Sex (patients)	
Male	5 (35.7)
Female	9 (64.3)
Smoking status	
Current or previous smoker	3 (21.4)
Never smoker	11 (78.6)
Histology (nodules)	
Primary lung cancer	6 (40.0)
Metastatic cancer	8 (53.3)
Inflammatory change	1 (6.7)
Location	
RUL	5 (33.3)
RLL	6 (40.0)
LUL	4 (26.7)
Nodule type	
Solid	10 (66.7)
Mixed	5 (33.3)
Nodule volume (mm ³)	706.9±653.6 [82.0–2,629.0]
Nodule size (maximum diameter mm)	11.1±4.9 [6.0–26.0]
Ablation time (min)	30.0±12.1 [1.3–30.0]
Post-cryoablation surgical resection (nodule)	
Yes	8 (53.3)
No	7 (46.7)
Post-cryoablation interval to surgical resection (day)	82±46 [27–137]
Cryoablation guidance approach	
EBUS	1 (7.1)
RAB and CBCT	14 (100)
Immunotherapy	
Yes	5 (35.7)
No	9 (64.3)

Data are presented as number (percentage) or mean ± SD [range]. CBCT, cone-beam computed tomography; EBUS, endobronchial ultrasound; LUL, left upper lobe; RLL, right lower lobe; RUL, right upper lobe.

pathology, disease stage, overall treatment intent, and patient-specific factors such as surgical fitness and comorbidities. Of the 15 treated nodules, 8 underwent subsequent surgical resection, while the remaining 7 were managed with radiologic surveillance. For patients who underwent surgical resection, operative details and histopathologic findings were extracted, including the presence of inflammation, fibrosis, necrosis, or residual tumor.

Statistical Analysis

The patient information and the nodule characteristics are described with frequency, percentage, mean ± standard deviation and range as appropriate. Between-group comparisons were conducted with paired Student *t* tests. The comparisons of the survival curve between groups were performed with the Kaplan-Meier log-rank test. In all analyses, a *p* value

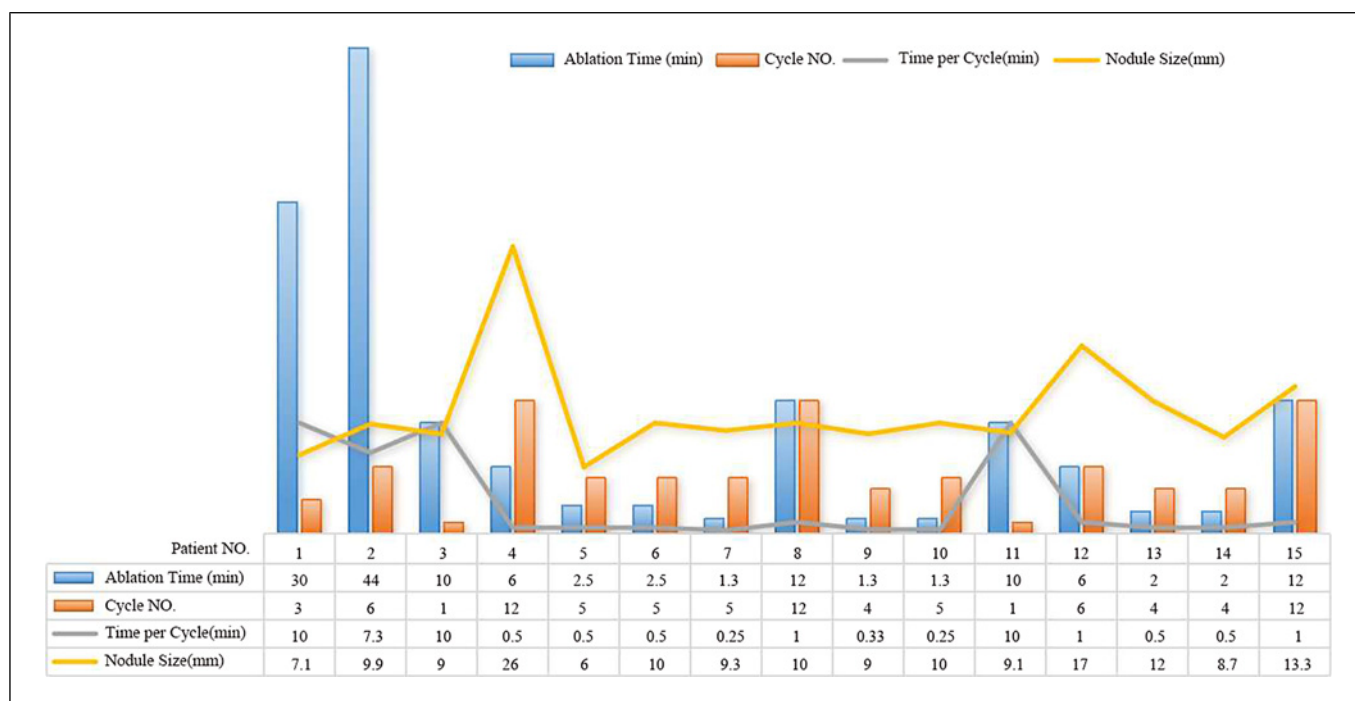


Fig. 1. Detail of the cryoablation time, cycle numbers, time per cycle, and the nodule size for the 15 patients. Patient number 1–8: surgical resection after cryoablation; patient number 9–15: radiologic surveillance.

less than 0.05 was considered significant. All statistical analyses were performed with GraphPad Prism (version 10, San Diego, CA, USA).

Results

Patient, Tumor, and Transbronchial Ablation Characteristics

The cohort comprised 14 patients presenting with a total of 15 lung nodules. Baseline patient and nodule characteristics are summarized in Table 1 and online supplementary Table 1 (for all online suppl. material, see <https://doi.org/10.1159/000551795>). The mean age of 70.3 ± 8.6 years; 5 patients were male. Histological diagnoses included 6 (40%) primary lung cancers, 8 (53.3%) metastatic lesions, and 1 (6.7%) with inflammatory change.

Nodules were located in the right upper lobe (5/15, 33.3%), right lower lobe (6/15, 40%), and left lower lobe (4/15, 26.7%). Most of the nodules were solid (10/15, 66.7%), and the rest were of mixed density. The mean pre-ablation nodule diameter was 11.1 ± 4.9 mm, and the mean volume was 706.9 ± 653.6 mm³. The mean total cryoablation time was 30.0 ± 12.1 min.

Eight nodules from 7 patients were subsequently resected after cryoablation (mean interval of 82 ± 46 days in-between). Figure 1 showed the detail of the cryoablation time, cycle numbers, time per cycle, and the nodule size for the 15 patients, among this, patient number from 1–8 were the patients who underwent surgical resection after cryoablation, and the other 7 patients (patient 9–15) underwent radiologic surveillance.

Safety and Feasibility

Transbronchial cryoablation was successfully completed in all 15 cases, yielding a technical feasibility rate of 100%. ssRAB navigation with CBCT and/or rEBUS confirmation provided accurate probe placement in all procedures. No aborted procedures occurred.

No mortality was reported within 60 days of the transbronchial cryoablation procedure. Adverse events including peri-procedural pneumothorax, peri-procedural pneumonia, bloody sputum/hemoptysis, pleural effusion, shortness of breath, chest pain/pleural pain were examined within 14 days of the procedure (Table 2). Three out of 14 (21.4%) patients reported procedure-related throat irritation, which is the most frequently reported minor adverse event. Two out of 14 (14.3%) patients reported shortness of breath after transbronchial ablation. One out 14 (7.1%)

Table 2. Adverse events of transbronchial cryoablation ($N = 14$)

Adverse Events	Frequency
Throat irritation	3 (21.4)
Shortness of breath	2 (14.3)
Cough	1 (7.1)
Pulmonary infection	1 (7.1)
Pleuritic pain	1 (7.1)
Minimal hemoptysis	1 (7.1)
Pneumothorax	0 (0)
Pleural effusion	0 (0)

Data are presented as number (percentage).

patients reported cough, minimal hemoptysis, pleuritic pain, or pulmonary infection. There was 1 patient who developed pulmonary infection, which was considered unrelated to cryoablation because the affected regions were anatomically distinct from the treatment site (the treated nodule was in the right upper lobe, whereas pneumonia developed in bilateral lower lobes). All events resolved with conservative management. Overall, the safety profile demonstrated low complication rates consistent with the expected profile of bronchoscopic cryotherapy.

Efficacy

The median follow-up time after the transbronchial cryoablation was 110 days, with a range of 13–487 days. The median local progression-free time was 56 days, with a range of 13–487 days (Table 1). During radiographic follow-up, 2 out of 15 nodules demonstrated local progression, resulting in radiographic local control in 86.7% of nodules (Fig. 2a, b).

The diameter of the nodule before and after transbronchial cryoablation did not yield a significant difference. More nodules (8 out of 15) increased in size on the first follow-up chest CT after cryoablation (performed 1–170 days post-procedure), as shown in Figure 2c (see online suppl. Table 1). Representative images of acute changes peri-transbronchial cryoablation demonstrated the nodule size increase, which was likely contributed by inflammation-related changes (Fig. 2d).

Cohort of Patients Undergoing Surgical Resection after Transbronchial Cryoablation

Seven patients (8 nodules) were followed up with CT scan (median 41.5 days) and then underwent surgical resection. The median interval between cryoablation and

resection was 79 days (range, 41–125) (Fig. 3a). Primary adenocarcinoma accounted for five of the resected nodules; three were metastatic fibrosarcoma. One nodule showed radiographic progression prior to surgery (Fig. 3b).

Histopathology examination of the surgically resected nodules, particularly primary lung adenocarcinomas, demonstrated marked post-cryoablation tissue changes, including prominent inflammation (Fig. 3c) and partial fibrosis (Fig. 3d), findings consistent with cryoablation-induced tissue injury (online suppl. Table 2). Compared with radiologic-surveillance cohort, the surgical cohort had a significantly shorter follow-up duration, as expected based on planned operative timeline.

Cohort of Patients Undergoing Regular Monitoring after Transbronchial Cryoablation

Seven patients (7 nodules) underwent imaging surveillance after ablation for a median of 358 days (range, 141–487). Among these, five were metastatic lung cancer nodules, one was an inflammatory nodule, and one was a primary lung cancer nodule in a patient medically unsuitable for surgery (online suppl. Table 1). Most nodules (6 out of 7, 85.7%) remained under control after transbronchial cryoablation, with one metastatic pancreatic cancer lesion showing progression at day 225 post-cryoablation (Fig. 3e, f), which occurred in the setting of rapid progression of the primary disease.

Serial CT scans demonstrated typical post-cryoablation evolution, including transient enlargement, cavitation, and gradual involution of the ablated zone (Fig. 4a, b). In the example shown in Figure 4a, cryoablated nodules demonstrated sustained stability during long-term follow-up, while new nodules and pleural effusion developed in other regions of the lung. Notably, a separate cryoablated nodule exhibited cavity formation over time (Fig. 4b), illustrating the efficacy of cryoablation in inducing local cell death. During follow-up, this cavity gradually diminished and was absorbed, with a corresponding reduction in nodule size.

Subgroup Efficacy Analysis

To further explore potential associations between ablation time or nodule characteristics and local progression, exploratory subgroup comparison was performed. Local progression rates were similar between the surgical and surveillance cohorts, despite longer surveillance duration in the latter (Fig. 5a). In this limited cohort, ablation time (long vs. short ablation time with a cutoff value of 5 min or 10 min) was not associated with local tumor progression (p value = 0.5614, Figure 5b; online suppl. Fig. 1); however, this analysis was exploratory and limited by the small

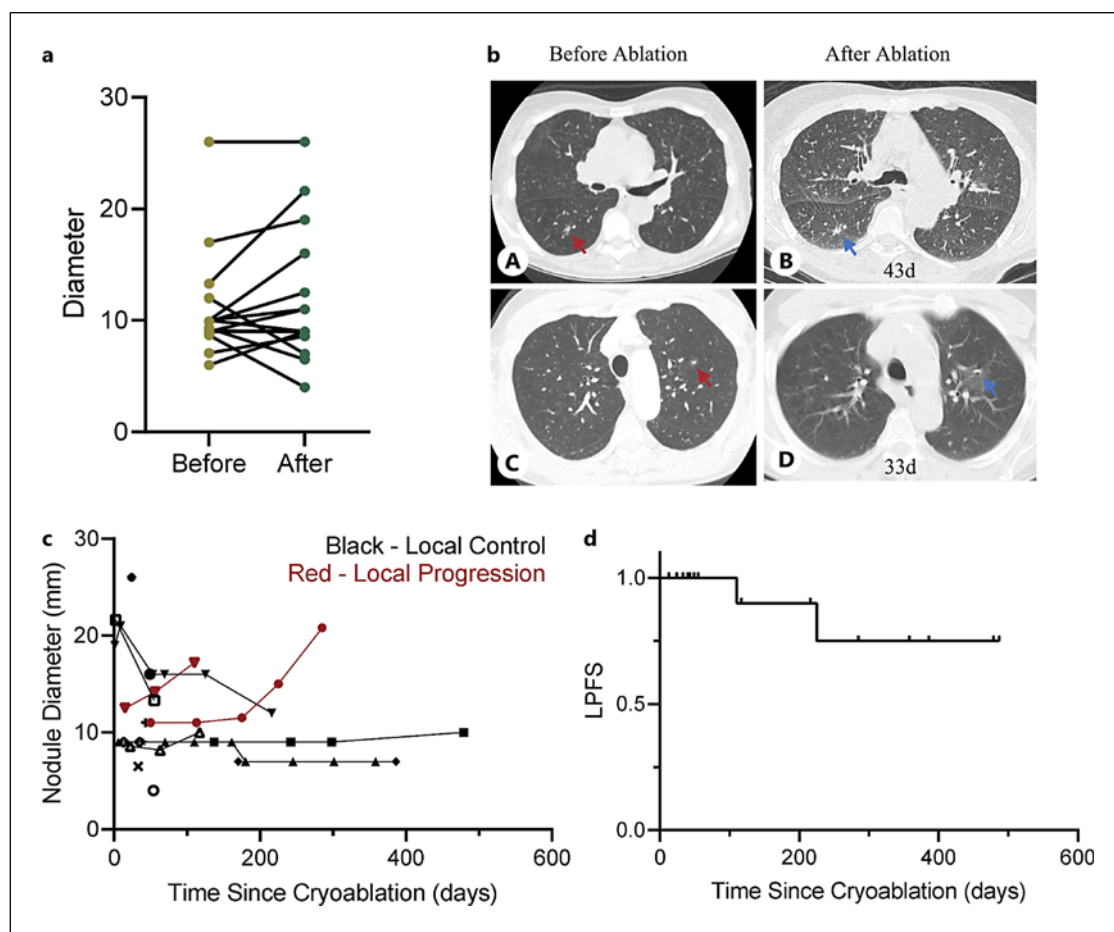


Fig. 2. Overview of all pulmonary lung nodule size changes with transbronchial cryoablation. **a** Pulmonary nodule diameter changes of all patients across serial follow-up CTs. **b** Kaplan-Meier curve of local progression-free survival of all pulmonary nodules after transbronchial cryoablation. **c** Measurement of

pulmonary nodule size before transbronchial cryoablation and at first CT follow-up after cryoablation. **d** CT images of two primary lung cancers before and after cryoablation. Panels (**a–d**), show the corresponding pre- (red arrows) and post-ablation (blue arrows) comparisons. LPFS, local progression-free survival.

number of progression events. Additionally, tumor pathology or histology did not demonstrate to be a factor for impacting tumor progression, with no significant difference observed between the nodules that were primary lung cancer or metastatic lesions (p value = 0.5685, Fig. 5c). Interestingly, patients who underwent immunotherapy as treatment for their primary cancer demonstrated a non-significant trend toward longer local progression-free survival (p value = 0.1154, Fig. 5d).

Discussion

This study provides one of the earliest clinical evaluations of ssRAB combined with CBCT to guide transbronchial cryoablation using a CO₂-based ERBE-

CRYO-2 platform. The findings demonstrate that this approach is technically feasible, has a favorable safety profile, and can achieve encouraging early local control in both primary and metastatic pulmonary nodules. The results contribute to the limited but growing evidence supporting bronchoscopic cryoablation as a potential minimally invasive treatment option for selected patients with pulmonary lesions [20]. Additionally, the results highlight cryoablation's potential as an adjunct therapy, consistent with NCCN guidelines that acknowledge image-guided thermal ablation [21] as an option for medically inoperable cases, supported by long-term follow-up data.

We achieved a 100% technical success rate, reflecting the advantages of ssRAB navigation and CBCT confirmation for precise probe placement [20]. Prior human

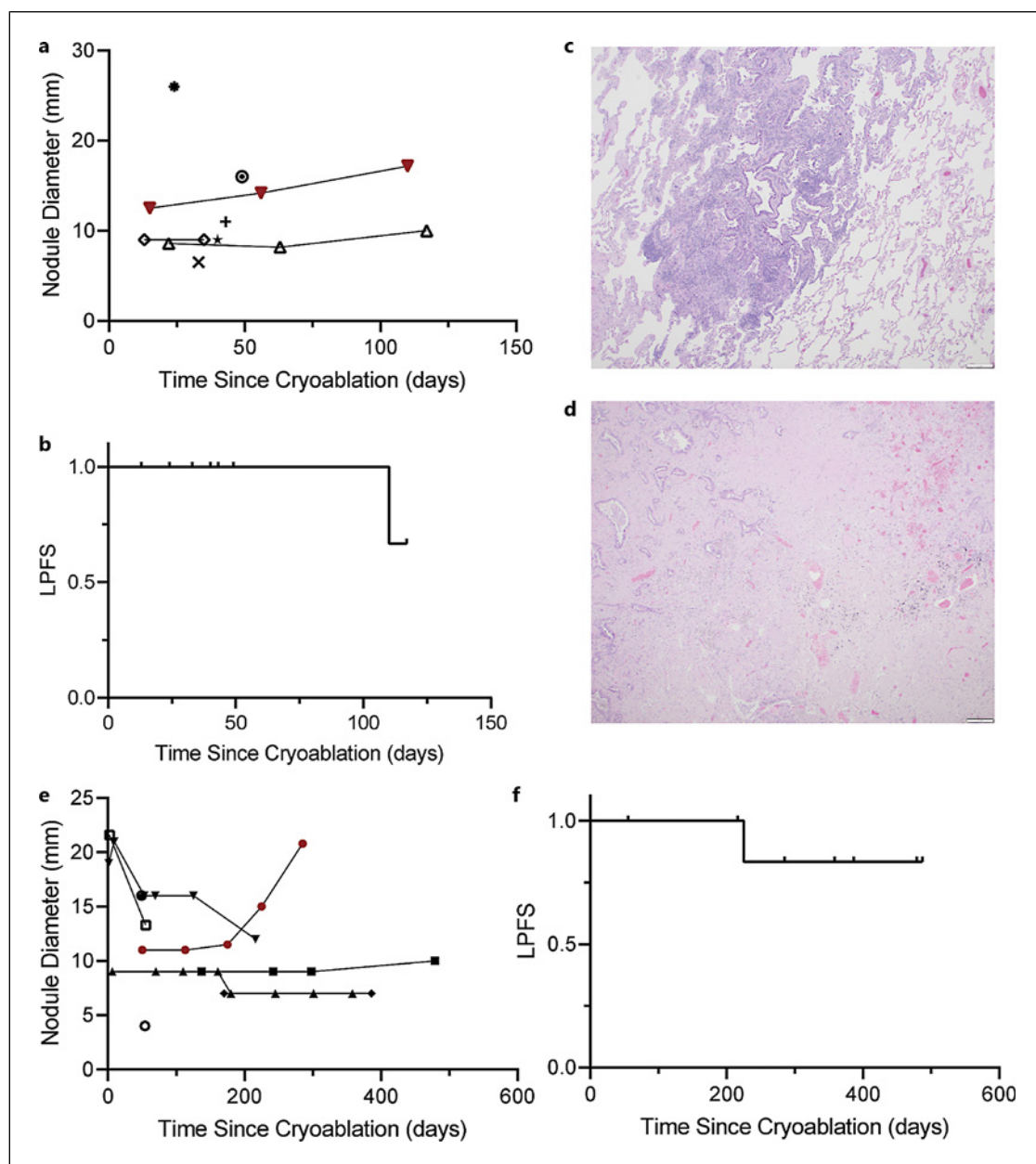


Fig. 3. Pulmonary lung nodule size changes and post-cryoablation histopathologic findings following transbronchial cryoablation. **a, b** Changes in pulmonary nodule diameter and local progression-free survival (LPFS) in patients who subsequently underwent surgical resection after cryoablation. **a** Serial changes in pulmonary nodule diameter on follow-up CT imaging. **b** Kaplan-Meier curve of LPFS in the surgical resection cohort. **c-d** Representative histopathologic findings from surgically re-

sected nodules following cryoablation, demonstrating post-ablation inflammation (**c**) and areas of partial fibrosis (**d**), consistent with cryoablation-induced tissue injury. **e-f** Changes in pulmonary nodule diameter and local progression-free survival (LPFS) in patients managed with radiologic surveillance after cryoablation. **e** Serial changes in pulmonary nodule diameter on follow-up CT imaging. **f** Kaplan-Meier curve of LPFS in the surveillance cohort. LPFS, local progression-free survival.

and swine studies of transbronchial or percutaneous cryoablation have predominantly utilized nitrogen- or argon-based cryoprobes, which generate larger and more

uniform ice balls than CO₂-based systems. [4, 10–12, 20, 22, 23]. In contrast, the ERBECRYO-2 platform has historically been used for central airway applications

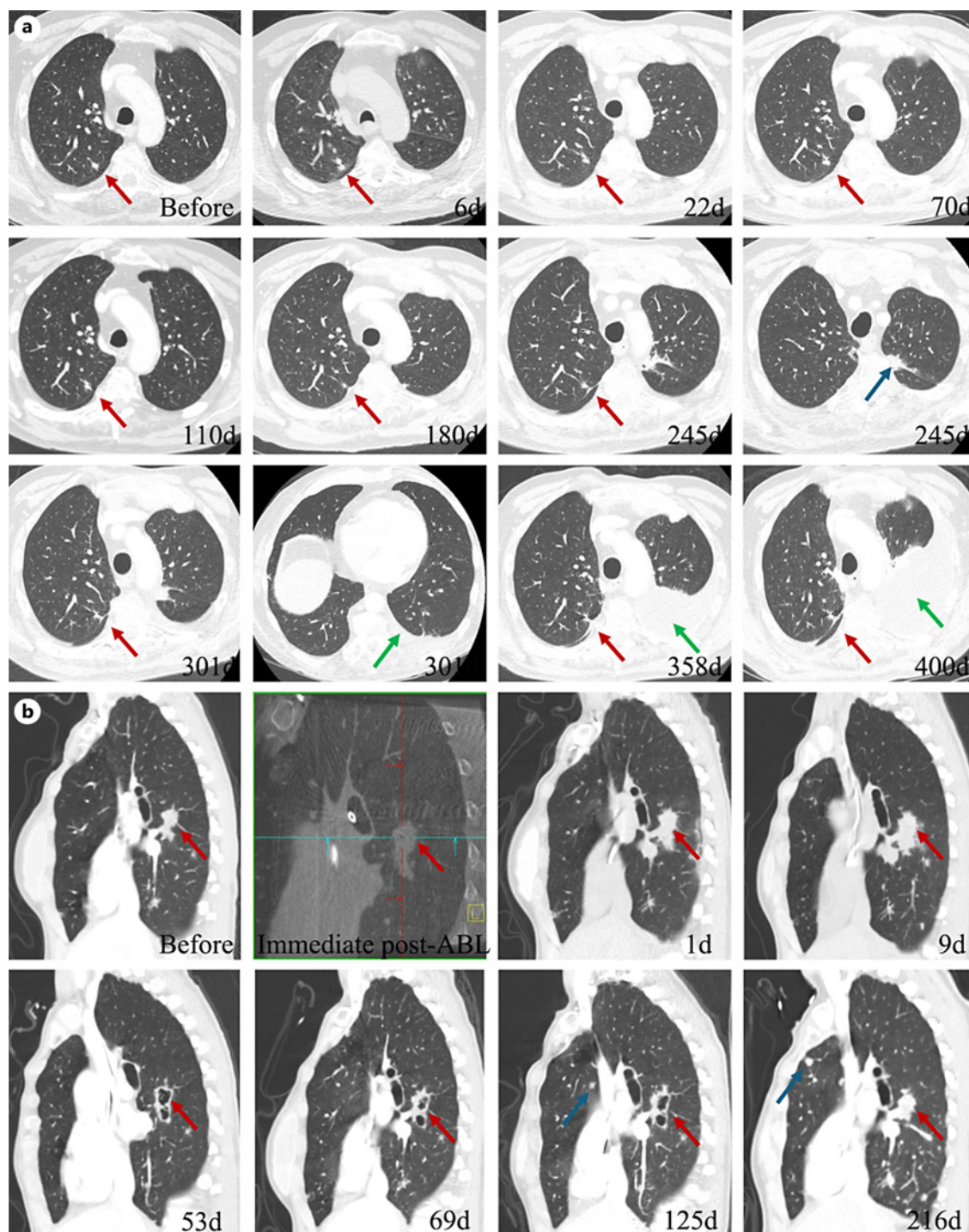


Fig. 4. Representative images of pulmonary lung nodule size changes in cohort of patients undergone regular monitoring after transbronchial cryoablation. **a** Representative images of metastatic lung lesions after cryoablation across sequential CT scans (red arrow) and newly developed metastatic lesion (blue arrow) and pleura effusion (green arrow) over time in a different lung

segment. **b** Follow-up chest CT images of a patient who underwent transbronchial cryoablation. The red arrow indicates a nodule treated with cryoablation, which initially increased in size and developed cavitation after ablation, followed by gradual resolution of the cavity and reduction in nodule size over time. The blue arrow indicates a newly emerged lesion during follow-up.

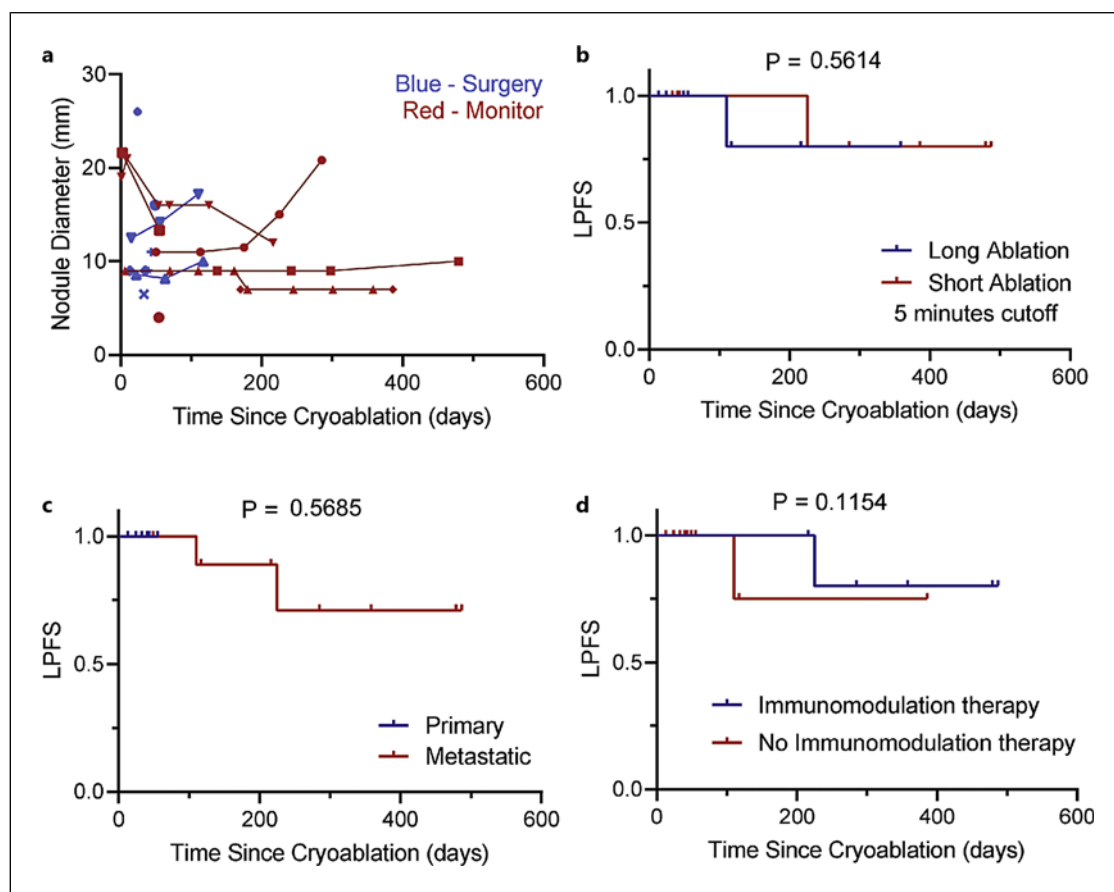


Fig. 5. Subgroup analysis of after transbronchial cryoablation. **a** Pulmonary nodule diameter changes of patients in the surgery group and regular monitoring group after cryoablation across serial follow-up CTs. **b–d** Local progression-free survival after transbronchial cryoablation, stratified by ablation time (**b**, cutoff value-5 min), tumor pathology (**c**), and immunomodulation therapy (**d**). LPFS, local progression-free survival.

such as cryotherapy, cryobiopsy, and tumor debulking rather than parenchymal ablation [17]. To date, there are no published clinical series evaluating peripheral lung nodule cryoablation using the ERBE system. The only relevant evidence is an in vitro pilot study, which demonstrated that a transbronchial cryoprobe based on the ERBE platform can generate ablative ice balls under controlled bench conditions [24]. From this practical perspective, extending a familiar bronchoscopic cryotherapy platform to parenchymal targets may represent a potential indication expansion and could, in theory, facilitate clinical implementation compared with the introduction of entirely new cryoablation systems; however, this interpretation remains speculative and requires validation in broader clinical experience.

Robotic bronchoscopy enhances procedural stability and allows seamless integration with real-time imaging, which may mitigate limitations associated with probe

flexibility and cryogen characteristics. The present results indicate that, despite the intrinsic differences between CO₂-based and nitrogen-based systems, the ER-BECRYO-2 probe can be delivered safely and effectively into peripheral lesions when paired with RAB and CBCT.

Safety profile is a crucial factor in guiding treatment plans for patients with pulmonary nodules. When surgical options are contraindicated or declined by patients, SBRT or percutaneous ablation are recommended by current clinical guidelines, with documented efficacy in achieving local tumor control [21]. In our study, there were no procedure-related serious adverse events needing medical intervention for robotic navigation-guided transbronchial cryoablation, demonstrating its favorable safety profile, consistent with prior reports of endobronchial cryotherapy. Importantly, unlike percutaneous approaches, transbronchial access avoids pleural

puncture, which may reduce the risk of pneumothorax and hemothorax – common complications of percutaneous cryoablation and microwave ablation reported in up to 60–70% of cases [25–28]. Although our cohort is modest in size, these findings support the premise that bronchoscopic cryoablation may offer a less invasive alternative for patients who are poor candidates for percutaneous ablation. Larger cohorts at multiple institutions with multiple operators are needed to confirm this observation.

Our longitudinal follow-up data provide encouraging evidence of the efficacious and durable local tumor control across a spectrum of lung nodules, including primary lung cancer and metastatic lung lesions. Overall local tumor control was 86.7% over a medium 110-day follow-up. Specifically, in metastatic lung cancer nodules, long-term follow-up consistently demonstrated stable control of the cryoablated lesions (up to 487 days), even in cases of primary disease progression. Limited by the number of patients in our study, ablation time or tumor characteristics (primary vs. metastatic lesions) did not yield significant association with long-term efficacy.

In our study, early post-ablation enlargement, cavitation, and inflammatory changes were frequently observed, consistent with the expected trajectory of cryoablation-induced injury and healing. Histopathologic examination of resected nodules further confirmed inflammation and fibrosis, supporting radiographic interpretations and aligning with preclinical cryoablation models [12, 29]. Cryoablation-induced cell death is primarily driven by intracellular ice crystal formation, osmotic injury, and membrane disruption during thaw, with repeated freeze-thaw cycles amplifying tissue lethality [30, 31]. These findings provide direct evidence supporting the multifaceted roles of cryoablation in inducing tumor necrosis, eliciting local inflammatory responses and fibrosis, which collectively contribute to effective local tumor control [14]. Importantly, in exploratory subgroup analysis, patients who received systemic immunotherapy demonstrated a nonsignificant trend toward longer local progression-free survival compared with those who did not receive immunotherapy. This observation should be interpreted with caution given the limited sample size and lack of statistical significance. The hypothesis that local immune responses induced by cryoablation may interact with systemic immunotherapies has been proposed [3]. Cryoablation-induced cellular disruption has been shown to release immunogenic components, including chemokines, cytokines, nucleic acids, and membrane fragments, which may promote antigen presentation and

T-cell activation – a phenomenon historically described as a cryo-immunologic response (CIR) and reported to be enhanced by combinatorial strategies in preclinical studies [3, 31]. Recent first-in-human studies of bronchoscopic cryoimmunotherapy have also demonstrated systemic immune activation, including changes in inflammatory biomarkers and T-cell populations, suggesting the presence of a clinically measurable cryo-immunologic effect in lung cancer patients [16]. Whether such immunologic interactions translate into meaningful clinical benefit when cryoablation is combined with immunotherapy remains uncertain and warrants evaluation in larger, prospective studies.

While our study provides valuable evidence regarding the safety and feasibility of robotic navigation-guided transbronchial cryoablation, it has several limitations. First, the number of nodules/patients included in this study is relatively small, with 15 nodules in 14 patients. This limits multivariable analysis to identify associated factors such as nodule histopathology and ablation time that may impact the efficacy of transbronchial cryoablation. Expanding patient cohorts and extending follow-up duration are necessary to address this limitation. Second, the retrospective design introduces potential selection bias and variability in imaging follow-up intervals. Third, results from this single-center study have limited generalization potential. Large-scale, multiple-center studies are needed to validate and generalize these results across broader patient populations and clinical settings. Fourth, robotic bronchoscopy-assisted transbronchial cryoablation requires experienced practitioners, underscoring the importance of formal and comprehensive training programs. Fifth, standardized criteria for radiographic assessment of bronchoscopic cryoablation zones are not yet established, and inflammatory changes may confound early post-ablation interpretation. Finally, the optimal freeze-thaw parameters, number of cycles, and real-time intraprocedural markers of complete ablation remain undefined for the ERBECRYO-2 platform.

Despite these limitations, our study provides preliminary evidence of the feasibility and safety of robotic navigation-guided transbronchial cryoablation in the management of pulmonary nodules and clinical evidence of its long-term efficacy. Future studies should incorporate prospective designs, standardized imaging protocols, and comparative cohorts to better define the parameters that optimize ablation efficacy and durability. Pathologic correlation – particularly in early-phase studies – will remain essential to guiding protocol refinement.

Conclusions

In this single-center retrospective study, ssRAB with CBCT guidance enabled accurate and reliable delivery of CO₂-based transbronchial cryoablation for primary and metastatic pulmonary nodules. The technique demonstrated excellent technical feasibility, a favorable safety profile, and encouraging early local control, including in patients with metastatic disease.

These findings support the potential role of bronchoscopic cryoablation as a minimally invasive therapeutic option, particularly for patients who are medically inoperable or unsuitable for standard therapeutic modalities, such as SBRT or percutaneous ablation. Potential integration with systemic therapies – especially immunotherapy – merits further exploration given the biological rationale for synergistic immune activation. Prospective multicenter studies with standardized cryoablation protocols, longer follow-up, and comparative effectiveness analyses are needed to further define the role of transbronchial cryoablation and establish its place within multidisciplinary lung cancer care.

Statement of Ethics

The study was conducted according to the Declaration of Helsinki (revised in 2013). Written informed consent was obtained from patients before the transbronchial cryoablation procedure, and approval was obtained from the UC San Diego Institutional Review Board (Approval No. #812977).

References

- 1 Sidorenkov G, Stadhouders R, Jacobs C, Mohamed Hoesin FAA, Gietema HA, Nackaerts K, et al. Multi-source data approach for personalized outcome prediction in lung cancer screening: update from the NELSON trial. *Eur J Epidemiol.* 2023;38(4):445–54. <https://doi.org/10.1007/s10654-023-00975-9>
- 2 Loverdos K, Fotiadis A, Kontogianni C, Iliopoulou M, Gaga M. Lung nodules: a comprehensive review on current approach and management. *Ann Thorac Med.* 2019; 14(4):226–38. https://doi.org/10.4103/atm.ATM_110_19
- 3 Laeseke P, Ng C, Ferko N, Naghi A, Wright GWJ, Zhang Y, et al. Stereotactic body radiation therapy and thermal ablation for treatment of NSCLC: a systematic literature review and meta-analysis. *Lung Cancer.* 2023;182:107259. <https://doi.org/10.1016/j.lungcan.2023.107259>
- 4 Castillo-Fortuno A, Paez-Carpio A, Matute-Gonzalez M, Odisio EG, Vollmer I, Baetens T,

- et al. Lung cryoablation: patient selection, techniques, and postablation imaging. *RadioGraphics.* 2025;45(6):e240157. <https://doi.org/10.1148/rg.240157>
- 5 Pusceddu C, Sotgia B, Fele RM, Melis L. CT-guided thin needles percutaneous cryoablation (PCA) in patients with primary and secondary lung tumors: a preliminary experience. *Eur J Radiol.* 2013;82(5):e246–53. <https://doi.org/10.1016/j.ejrad.2012.12.010>
- 6 Zhong C, Chen E, Su Z, Chen D, Wang F, Wang X, et al. Safety and efficacy of a novel transbronchial radiofrequency ablation system for lung tumours: one year follow-up from the first multi-centre large-scale clinical trial (BRONC-RFII). *Respirology.* 2025;30(1): 51–61. <https://doi.org/10.1111/resp.14822>
- 7 Lau KKW, Lau RWH, Baranowski R, Krzykowski J, Ng CSH. Transbronchial microwave ablation of peripheral lung tumors: the NAVABlate study. *J Bronchology Interv*

- Pulmonol.* 2024;31(2):165–74. <https://doi.org/10.1097/LBR.0000000000000950>
- 8 Chan JWY, Lau RWH, Ngai JCL, Tsoi C, Chu CM, Mok TSK, et al. Transbronchial microwave ablation of lung nodules with electromagnetic navigation bronchoscopy guidance—a novel technique and initial experience with 30 cases. *Transl Lung Cancer Res.* 2021;10(4):1608–22. <https://doi.org/10.21037/tlcr-20-1231>
- 9 Tang Y, Tian S, Chen H, Li X, Pu X, Zhang X, et al. Transbronchial lung cryobiopsy for peripheral pulmonary lesions. A narrative review. *Pulmonology.* 2024;30(5):475–84. <https://doi.org/10.1016/j.pulmoe.2023.08.010>
- 10 Gu C, Yuan H, Yang C, Xie F, Chen J, Zhu L, et al. Transbronchial cryoablation in peripheral lung parenchyma with a novel thin cryoprobe and initial clinical testing. *Thorax.* 2024;79(7): 633–43. <https://doi.org/10.1136/thorax-2023-220227>

Conflict of Interest Statement

Keriann Van Nostrand – Speaker payments for/from Pinnacle Biologics, Intuitive Surgical, Wolf Medicine, and Biodesix. Jorge A. Munoz Pineda – Research support from Ruby Robotics and consulting payments from Intuitive and Richard Wolf. Niraj Patel – consulting for Intuitive Surgical, Olympus, and LeadOptik. George Cheng – consulting relationships with Ajax and Serpex, and speaker honoraria from Intuitive Surgical, Olympus, Cook, Pinnacle Biologics, Siemens, and Wolfe. Felix J. Herth was a member of the journal's Editorial Board at the time of submission. The remaining authors have no conflicts of interest to declare.

Funding Sources

This study was not supported by any sponsor or funder.

Author Contributions

(I) Conception and design: L.X., N.P., and G.C.; (II) administrative support: all authors; (III) provision of study materials or patients: K.V.N., N.P., B.D., J.A.M.P., R.M., and G.C.; (IV) collection and assembly of data: L.X., W.G., B.T., and G.C.; (V) data analysis and interpretation: L.X., W.G., M.Z., G.L., and G.C.; (VI) manuscript writing and editing: L.X., W.G., K.V.N., N.P., J.A.M.P., F.H., and G.C.; (VII) final approval of manuscript: all authors.

Data Availability Statement

The data included in this manuscript are available upon reasonable request and Ethics Committee review. Further inquiries can be directed to the corresponding author.

- 11 de Ruiter QMB, Mauda-Havakuk MM, Starost MF, Bakhtashvili I, Esparza-Trujillo JA, Brown A, et al. Image-guided trans-bronchial pulmonary cryoablation with a flexible cryoprobe in swine: performance and radiology-pathology correlation. *J Vasc Interv Radiol.* 2024;35(7):1022–30.e4. <https://doi.org/10.1016/j.jvir.2024.02.026>
- 12 Zheng X, Yuan H, Gu C, Yang C, Xie F, Zhang X, et al. Transbronchial lung parenchyma cryoablation with a novel flexible cryoprobe in an in vivo porcine model. *Diagn Interv Imaging.* 2022;103(1):49–57. <https://doi.org/10.1016/j.diii.2021.08.005>
- 13 Kohno M, Hashimoto R, Oiwa K, Yashiro H, Nakatsuka S, Kawamura M, et al. Initial experience with transbronchial cryoablation as a novel local treatment for malignant peripheral lung lesions. *BMJ Open Respir Res.* 2018;5(1):e000315. <https://doi.org/10.1136/bmjresp-2018-000315>
- 14 Velez A, DeMaio A, Sterman D. Cryoablation and immunity in non-small cell lung cancer: a new era of cryo-immunotherapy. *Front Immunol.* 2023;14:1203539. <https://doi.org/10.3389/fimmu.2023.1203539>
- 15 Hamakawa M, Niwa T, Fukuda Y, Yokoyama T, Ishida T. Transbronchial cryoablation as local treatment for central airway malignant tumor. *Respiration.* 2024;103(9):587–92. <https://doi.org/10.1159/000540299>
- 16 Tsay JCJ, Velez A, Collazo D, Laniado I, Bessich J, Murthy V, et al. A phase I dose-escalation clinical trial of bronchoscopic cryoimmunotherapy in advanced-stage NSCLC. *JTO Clin Res Rep.* 2025;6(8):100849. <https://doi.org/10.1016/j.jtocrr.2025.100849>
- 17 DiBardino DM, Lanfranco AR, Haas AR. Bronchoscopic cryotherapy: clinical applications of the cryoprobe, cryospray, and cryoadhesion. *Ann Am Thorac Soc.* 2016;13(8):1405–15. <https://doi.org/10.1513/AnnalsATS.201601-062FR>
- 18 Criner GJ, Eberhardt R, Fernandez-Bussy S, Gompelmann D, Maldonado F, Patel N, et al. Interventional bronchoscopy. *Am J Respir Crit Care Med.* 2020;202(1):29–50. <https://doi.org/10.1164/rccm.201907-1292SO>
- 19 Chen AC, Pastis NJ Jr, Mahajan AK, Khandhar SJ, Simoff MJ, Machuzak MS, et al. Robotic bronchoscopy for peripheral pulmonary lesions: a multicenter pilot and feasibility study (BENEFIT). *Chest.* 2021;159(2):845–52. <https://doi.org/10.1016/j.chest.2020.08.2047>
- 20 Zhang C, Xie F, Xi H, Liu S, Yan L, Wang Z, et al. Shape-sensing robotic-assisted bronchoscopy combined with cone-beam computed tomography-guided cryoablation for malignant lung tumors. *Respiration.* 2025;104(12):963–73. <https://doi.org/10.1159/000547904>
- 21 National Comprehensive Cancer Network. NCCN clinical practice guidelines in oncology: non-small cell lung cancer: NCCN; 2025. Available from: <https://www.nccn.org/guidelines> (accessed November 4, 2025).
- 22 Nomori H, Yamazaki I, Shiraishi A, Adachi T, Kanno M. Cryoablation for T1N0M0 non-small cell lung cancer using liquid nitrogen. *Eur J Radiol.* 2020;133:109334. <https://doi.org/10.1016/j.ejrad.2020.109334>
- 23 Páez-Carpio A, Gómez FM, Isus Olivé G, Paredes P, Baetens T, Carrero E, et al. Image-guided percutaneous ablation for the treatment of lung malignancies: current state of the art. *Insights Imaging.* 2021;12(1):57. <https://doi.org/10.1186/s13244-021-00997-5>
- 24 Hammer D, Büdi L, Nagy Á, Varga R, Horváth P. Evaluation of a transbronchial cryoprobe for the ablation of pulmonary nodules: an in vitro pilot study. *BMC Pulm Med.* 2023;23(1):71. <https://doi.org/10.1186/s12890-023-02358-y>
- 25 Kaseda K, Asakura K, Nishida R, Okubo Y, Masai K, Hishida T, et al. Feasibility and safety of percutaneous cryoablation under local anesthesia for the treatment of malignant lung tumors: a retrospective cohort study. *J Thorac Dis.* 2022;14(11):4297–308. <https://doi.org/10.21037/jtd-22-830>
- 26 Inoue M, Nakatsuka S, Yashiro H, Ito N, Izumi Y, Yamauchi Y, et al. Percutaneous cryoablation of lung tumors: feasibility and safety. *J Vasc Interv Radiol.* 2012;23(3):295–305; quiz 305. <https://doi.org/10.1016/j.jvir.2011.11.019>
- 27 Xu Z, Wang X, Ke H, Lyu G. Cryoablation is superior to radiofrequency ablation for the treatment of non-small cell lung cancer: a meta-analysis. *Cryobiology.* 2023;112:104560. <https://doi.org/10.1016/j.cryobiol.2023.104560>
- 28 Abrishami Kashani M, Murphy MC, Saenger JA, Wrobel MM, Tahir I, Mrah S, et al. Risk of persistent air leaks following percutaneous cryoablation and microwave ablation of peripheral lung tumors. *Eur Radiol.* 2023;33(8):5740–51. <https://doi.org/10.1007/s00330-023-09499-y>
- 29 Gu C, Wang X, Wang K, Xie F, Chen L, Ji H, et al. Cryoablation triggers type I interferon-dependent antitumor immunity and potentiates immunotherapy efficacy in lung cancer. *J Immunother Cancer.* 2024;12(1):e008386. <https://doi.org/10.1136/jitc-2023-008386>
- 30 Erinjeri JP, Clark TW. Cryoablation: mechanism of action and devices. *J Vasc Interv Radiol.* 2010;21(8 Suppl):S187–91. <https://doi.org/10.1016/j.jvir.2009.12.403>
- 31 Baust JG, Snyder KK, Santucci KL, Robilotto AT, Van Buskirk RG, Baust JM. Cryoablation: physical and molecular basis with putative immunological consequences. *Int J Hyperthermia.* 2019;36(Suppl 1):10–6. <https://doi.org/10.1080/02656736.2019.1647355>